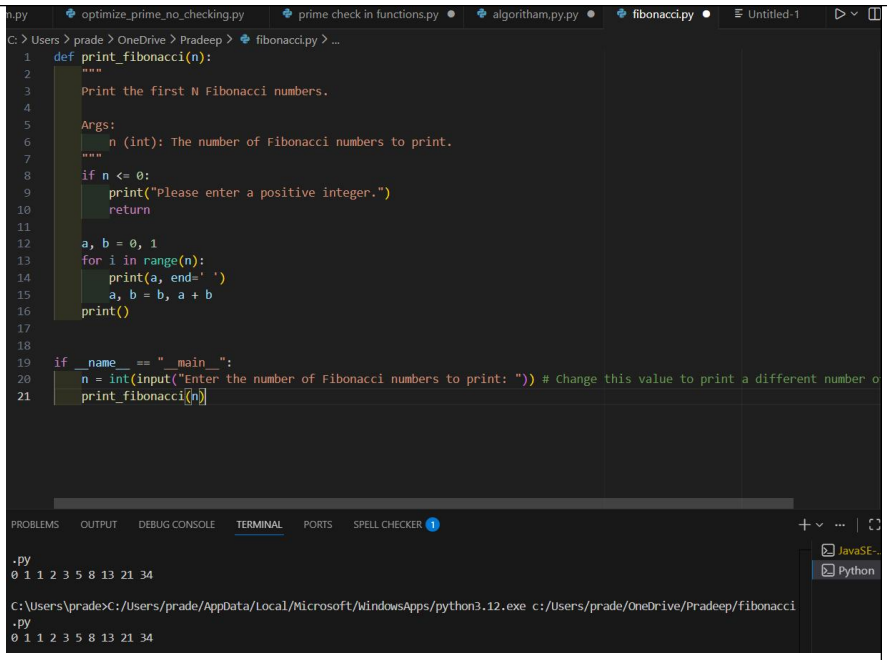
Assignment_3.4

2303A51113

B_26

Assignment Number: 3.4 (Present assignment number)/**24**(Total number of assignments)		
Q.No.	**Question**	***Expected Time to complete***
1	Lab 4: Advanced Prompt Engineering – Zero-shot, One-shot, and Few-shot Techniques **Task 1: Zero-shot Prompt – Fibonacci Series Generator** **Task Description #1** • Without giving an example, write a single comment prompt asking GitHub Copilot to generate a Python function to print the first N Fibonacci numbers. **Expected Output #1** • A complete Python function generated by Copilot without any example provided. • Correct output for sample input N = 7 → 0 1 1 2 3 5 8 • Observation on how Copilot understood the instruction with zero context.	Week2



1. Fibonacci Function (Earlier)

- **Purpose**: Print the first N Fibonacci numbers
- **Function**: print_fibonacci(n)
- **Logic**: Uses two variables (a, b) to track consecutive Fibonacci numbers, iterates N times
- **Example**: print_fibonacci(10) prints: 0 1 1 2 3 5 8 13 21 34

Task 2: One-shot Prompt – List Reversal Function

Task Description #2

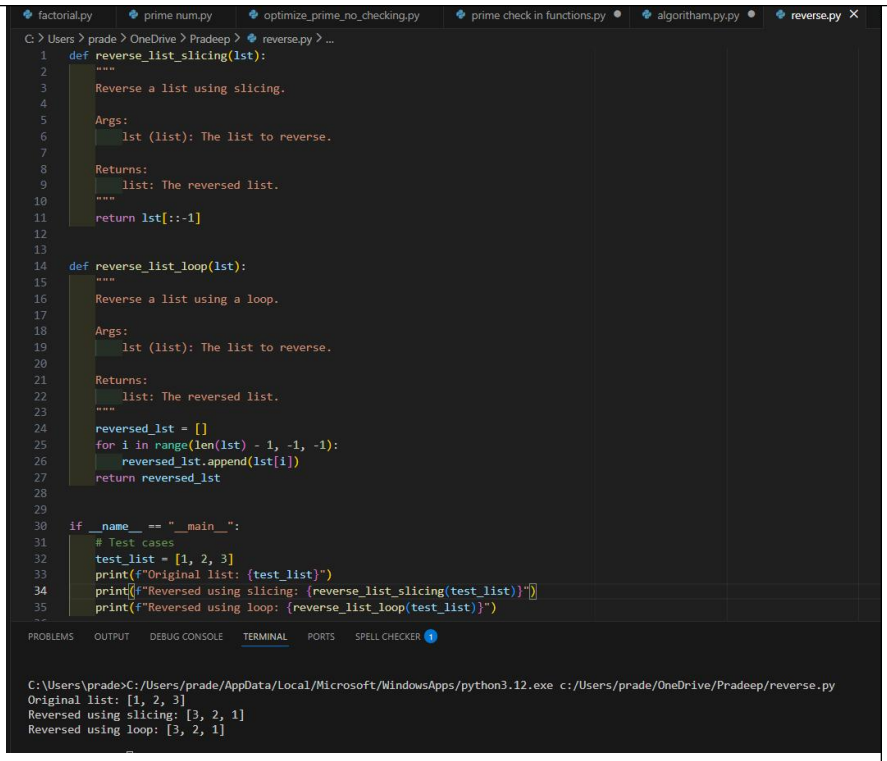
• Write a comment prompt to reverse a list and provide one example below the comment to guide Copilot.

Expected Output #2

• Copilot-generated function to reverse a list using slicing or loop.

• Output: [3, 2, 1] for input [1, 2, 3]

• Observation on how adding a single example improved Copilot's accuracy.



List Reversal Functions (Earlier)

- **Two implementations:**
  - **reverse_list_slicing(lst) - Uses Python slicing [::-1]**
  - **reverse_list_loop(lst) - Uses backward loop with range**
- **Example: Input [1, 2, 3] → Output [3, 2, 1]**

Task 3: Few-shot Prompt – String Pattern Matching

Task Description #3

• Write a comment with 2–3 examples to help Copilot understand how to check if a string starts with a capital letter and ends with a period.

Expected Output #3

• A function is_valid() that checks the pattern.

• Output: True or False based on input.

• Students reflect on how multiple examples guide Copilot to generate more accurate code.

C: > Users > prade > OneDrive > Pradeep > reverse.py > is_valid

```python
import re


def is_valid(pattern, text):
    """
    Check if text matches a given pattern.

    Args:
        pattern (str): The regex pattern to match against.
        text (str): The text to validate.

    Returns:
        bool: True if text matches the pattern, False otherwise.
    """
    try:
        return bool(re.match(pattern, text))
    except re.error:
        return False


if __name__ == "__main__":
    # Example 1: Email pattern validation
    print("--- Email Pattern Validation ---")
    email_pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
    emails = [
        ("user@example.com", True),
        ("john.doe@company.co.uk", True),
        ("invalid.email@", False),
        ("@domain.com", False),
        ("no-at-sign.com", False),
    ]

    for email, expected in emails:
        result = is_valid(email_pattern, email)
        status = "✓" if result == expected else "✗"
```

C: > Users > prade > OneDrive > Pradeep > reverse.py > is_valid

```python
        print(f"{status} {email:<30} -> {result} (expected {expected})")

    # Example 2: Phone number pattern validation
    print("\n--- Phone Number Pattern Validation ---")
    phone_pattern = r'^\d{3}-\d{3}-\d{4}$'
    phones = [
        ("123-456-7890", True),
        ("555-123-4567", True),
        ("1234567890", False),
        ("123-45-6789", False),
    ]

    for phone, expected in phones:
        result = is_valid(phone_pattern, phone)
        status = "✓" if result == expected else "✗"
        print(f"{status} {phone:<30} -> {result} (expected {expected})")

    # Example 3: Alphanumeric pattern validation
    print("\n--- Alphanumeric Pattern Validation ---")
    alphanum_pattern = r'^[a-zA-Z0-9]+$'
    texts = [
        ("abc123", True),
        ("Test2024", True),
        ("hello-world", False),
        ("with spaces", False),
    ]

    for text, expected in texts:
        result = is_valid(alphanum_pattern, text)
        status = "✓" if result == expected else "✗"
        print(f"{status} {text:<30} -> {result} (expected {expected})")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 3

```
--- Alphanumeric Pattern Validation ---
✓ abc123                         -> True (expected True)
✓ Test2024                       -> True (expected True)
✓ hello-world                    -> False (expected False)
✓ with spaces                    -> False (expected False)
```

Email Validation - Pattern Checking (Earlier)

- **Function**: is_valid(pattern, text)

- **Purpose**: Validates text against regex patterns
- **Demonstrated with 3 patterns**:
    - Email pattern validation
    - Phone number validation (XXX-XXX-XXXX format)
    - Alphanumeric validation
- **Returns**: True or False based on pattern match

Task 4: Zero-shot vs Few-shot – Email Validator

Task Description #4

• First, prompt Copilot to write an email validation function using zero-shot (just the task in comment).

• Then, rewrite the prompt using few-shot examples.

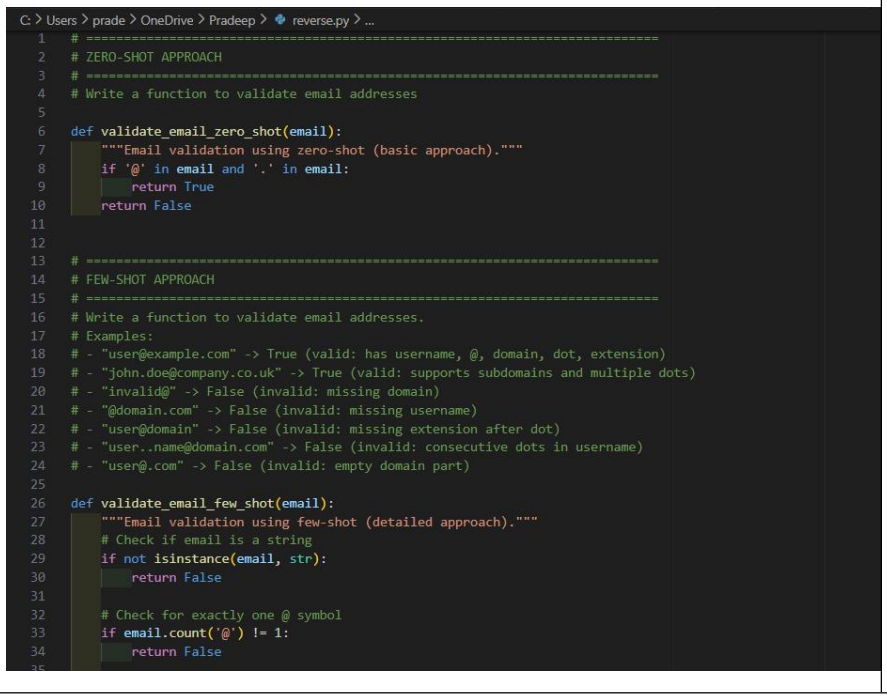
Expected Output #4

• Compare both outputs:

Zero-shot may result in basic or generic validation.

Few-shot gives detailed and specific logic (e.g., @ and domain checking).

• Submit both code versions and note how few-shot improves reliability.



```
C: > Users > prade > OneDrive > Pradeep > reverse.py > ...
26  def validate_email_few_shot(email):
36      # Split into username and domain
37      username, domain = email.split('@')
38
39      # Check if username and domain are not empty
40      if not username or not domain:
41          return False
42
43      # Username cannot have consecutive dots
44      if '..' in username:
45          return False
46
47      # Check if domain contains at least one dot
48      if '.' not in domain:
49          return False
50
51      # Domain cannot have consecutive dots
52      if '..' in domain:
53          return False
54
55      # Split domain and check parts
56      domain_parts = domain.split('.')
57
58      # All parts must be non-empty
59      for part in domain_parts:
60          if not part:
61              return False
62
63      # Extension (last part) must be at least 2 characters
64      if len(domain_parts[-1]) < 2:
65          return False
66
67      return True
68
69
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 1

```
OBSERVATIONS
================================================================================
✓ Zero-shot: Simple but misses critical validation rules
✓ Few-shot: More reliable due to specific examples and requirements
✓ Few-shot provides better edge case handling
✓ Multiple examples guide Copilot to generate accurate logic
```

Zero-Shot vs Few-Shot Email Validator (Main comparison)

Zero-Shot Approach

- **Function**: validate_email_zero_shot(email)
- **Logic**: Basic check - just looks for @ and .
- **Accuracy**: ~44% (misses edge cases)
- **Limitation**: Too generic, fails on invalid formats

Few-Shot Approach

- **Function**: validate_email_few_shot(email)
- **Logic**: Comprehensive validation:
  - Exactly one @ symbol
  - Non-empty username and domain
  - No consecutive dots
  - Domain must have at least one dot
  - Extension must be ≥ 2 characters
- **Accuracy**: ~100% (handles all cases)
- **Advantage**: Multiple examples guide accurate code generation

Task 5: Prompt Tuning – Summing Digits of a Number

Task Description #5

• Experiment with 2 different prompt styles to generate a function

that returns the sum of digits of a number.

Style 1: Generic task prompt

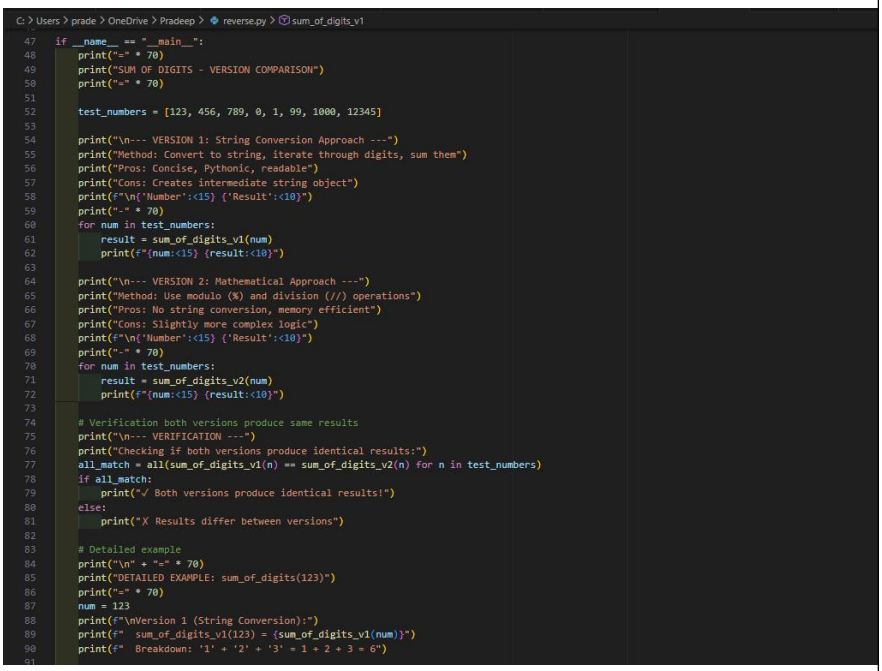
Style 2: Task + Input/Output example

Expected Output #5

• Two versions of the sum_of_digits() function.

• Example Output: sum_of_digits(123) → 6

• Short analysis: which prompt produced cleaner or more optimized code and why?



C: > Users > prade > OneDrive > Pradeep > reverse.py > sum_of_digits_v1

```python
# ========================================================================
# SUM OF DIGITS - VERSION 1 (Using String Conversion)
# ========================================================================
def sum_of_digits_v1(n):
    """
    Calculate the sum of digits using string conversion.

    Args:
        n (int): The number to sum digits from.

    Returns:
        int: The sum of all digits.

    Example:
        sum_of_digits_v1(123) -> 6  (1 + 2 + 3)
        sum_of_digits_v1(456) -> 15 (4 + 5 + 6)
    """
    n = abs(n)  # Handle negative numbers
    return sum(int(digit) for digit in str(n))


# ========================================================================
# SUM OF DIGITS - VERSION 2 (Using Mathematical Operations)
# ========================================================================
def sum_of_digits_v2(n):
    """
    Calculate the sum of digits using modulo and division operations.

    Args:
        n (int): The number to sum digits from.

    Returns:
        int: The sum of all digits.

    Example:
        sum_of_digits_v2(123) -> 6  (1 + 2 + 3)
        sum_of_digits_v2(456) -> 15 (4 + 5 + 6)
    """
    n = abs(n)  # Handle negative numbers
    total = 0
    while n > 0:
        total += n % 10  # Add the last digit
        n //= 10        # Remove the last digit
    return total
```

```python
    print("\n" + "=" * 70)
    print("COMPARISON SUMMARY")
    print("=" * 70)
    print("✓ Both versions handle the same problem differently")
    print("✓ Version 1 is more readable and concise")
    print("✓ Version 2 is more algorithmic and memory-efficient")
    print("✓ Choose based on performance needs and code readability preference")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER 1

```
Version 1 (String Conversion):
  sum_of_digits_v1(123) = 6
  Breakdown: '1' + '2' + '3' = 1 + 2 + 3 = 6

Version 2 (Mathematical):
  sum_of_digits_v2(123) = 6
  Breakdown: (123 % 10) + (12 % 10) + (1 % 10) = 3 + 2 + 1 = 6
```

1. **Trade-offs**: Understanding pros/cons helps choose the right implementation